

Now suppose that both "If you have a current password, then you can log onto the network." and "You have a current password." are true statements. Whenever we replace p by "You have a current password" and q by "You can log onto the network," it necessarily follows that you can log onto the network is true. This argument is valid because its form is valid. Note that whenever we replace ~~propositions~~ p and q by propositions where $p \rightarrow q$ and p are both true, then q must also be true.

What happens when we replace p and q in this argument form by propositions where not both p and $p \rightarrow q$ are true? For example, suppose that p represents "You have access to the network." and q represents "You can change your grade" and that p is true, but $p \rightarrow q$ is False. The argument we obtain by substituting these values of p and q into the argument form is

"If you have access to the network, then you can change your grade."
"You have access to the network."

$\therefore$ "You can change your grade."

The argument we obtained is a valid argument, but because one of the premises, namely the first premise, is false, we cannot conclude that the conclusion is true. (Most likely, this conclusion is false).

Definition 1: An argument in propositional logic is a sequence of propositions. All but the final proposition in the argument are called premises and the final proposition is called the conclusion. An argument is valid if the truth of all its premises implies that the conclusion is true.

$\rightarrow$ (Oshadharon idea)
An argument form in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is valid if no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.